CLAYTON GROTH

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PROFESSIONAL SUMMARY

Software Engineer with 7+ years of production experience across the full stack in Python, NodeJS, Go, Shell, and much more. Experienced building and maintaining production systems end-to-end: fullstack dev, backend services, cloud infrastructure, data pipelines, database design, DevOps, security, and more. Aside from the raw technical skills, my core assets as an engineer are deep curiosity, excellent problem solving skills, standout communication skills, and *conceptual* mastery of the entire stack. These traits make me excel at complex debugging on the terminal, writing features alone and in collaboration with a team (with and without AI tools), understanding and designing dense code, architectural whiteboard discussions, and managing other engineers to deliver production-ready features. I work daily in agentic AI-assisted development workflows. I know when these tools are most useful semi-autonomously and when something requires immediate hands-on expertise. As a leader, I lead by example and always put people first. To me this means fostering skill development and responsibility through curiosity and collaboration, tailored to the team and individuals I am leading. At my most recent role, I have ramped up and mentored most of the engineering team. I care about building and shipping exemplary software that holds up in production.

TECHNICAL SKILLS

Languages: Python (expert), JavaScript/TypeScript (expert), SQL (expert), Shell/Bash, GoLang, C++, Java, Rust

Debugging/Terminal: Linux/Unix terminal expertise, excellent problem solving under pressure, experience debugging complex systems involving: message-queues, networking logs, and complex parallelized file-io jobs

Backend: Node.js/Express, Flask, Django, FastAPI, REST APIs, GraphQL, multithreaded/async service design, distributed systems, OAuth 2.0, SAML

Frontend: React (expert), Redux, TypeScript, React Native, Redux and state management tools, responsive web design

Databases/Data: PostgreSQL/PostGIS (expert), MongoDB, Oracle SQL, SQLite, DuckDB, Snowflake; schema design, query optimization, spatial indexing, experience building and maintaining data pipelines

Infrastructure: Docker, Kubernetes, Helm, Ansible, Azure, AWS, CI/CD pipelines, Redis, RabbitMQ

AI & Agentic Systems: MCP (Model Context Protocol) server design & maintenance, agentic coding workflows (Claude Code), LLM tool-calling/orchestration patterns

Security: CISSP training, SAST/DAST scanning, vulnerability tracking and mitigation

Data & ML: NumPy, Pandas, SciPy, scikit-learn, large-scale parallel data processing, time series analysis

Geospatial: PostGIS, GeoPandas, Shapely, Rasterio, OSMNX, Mapbox GL, GDAL/OGR, QGIS

EXPERIENCE

Staff Software Engineer — cQuant
Remote | *January 2024 – Present*

- Built a production-grade MCP server with full OAuth 2.0 integration, using AgentGateway giving clients a secure AI-accessible interface to the company's API, and continue to own its ongoing maintenance against evolving MCP standards.
- Designed and led the platform's security posture from SAST/DAST scanning to manually discovering and patching many vulnerabilities.
- Work daily in agentic AI-assisted development workflows (e.g., Claude Code), using tool-calling and multi-step agent workflows as a routine part of feature development, debugging, and code review, while maintaining full engineering *ownership and judgment* over the output.
- Drove huge improvements in critical API response times and helped achieve 99.99% (SLA) uptime for the application. Achieved by horizontal microservice scaling (NodeJS, Go), query optimization, idempotency/concurrency guards, rate limiting, migrating compute-heavy work out of the HTTP-request path into dedicated multithreaded microservices, and leading efforts to redesign large chunks of the database and cloud infrastructure.
- Ramped up every engineer the company has hired over the years, including the current staff DevOps engineer and several core senior engineers. Managed teams of up to four engineers (two senior, two junior) plus offshore resources.

Senior Software Engineer — cQuant
Remote | *January 2022 – January 2024*

- Built and owned a graph-based constraint engine for real-time editing of huge and complex interrelated energy and financial modeling configs. A graph-data-structure based design here was needed for the combinatorial explosion across interdependent config dimensions.
- Designed and built Go services for massively parallelized job orchestration and file I/O.
- Built and owned a geospatial renewable energy asset valuation application handling spatial analysis and data visualization for energy grid data, asset pricing, and portfolio decisions.
- Designed and built an async file I/O service to handle cloud operations processing 100GB+ batches of files used in quantitative modeling. Addressed a critical bottleneck where the API-server had been taking this responsibility, causing slow response times.
- Built out the DevOps infrastructure from Ansible-based provisioning through to Kubernetes and Helm, and own full deployment pipelines from CI/CD through production.
- Built and owned the platform's OAuth 2.0 and SAML authentication flows.
- Designed and managed PostgreSQL databases/schemas and ORMs to efficiently handle application data.

Software Engineer — cQuant
Remote | *January 2021 – January 2022*

- Joined in the early startup stage and brought the application to an acquisition-ready stage. Worked closely with the CTO over four years to rebuild it into a scalable, cloud-hosted platform on Azure. Now serving a significantly larger growing client base.
- Built and maintained a wide range of internal tools and automations such as documentation pipelines, cloud blob cleanup, and reporting tools. (C++, NodeJS, Linux, JavaScript, React, Go, Python)
- Extensive terminal work debugging and monitoring production systems. Comfortable diagnosing issues under pressure.

Full Stack Developer — Digital Locale
Remote | *June 2019 – January 2021*

- Built a travel and mapping platform integrating multiple geospatial data sources with React and Mapbox GL, including REST APIs and PostGIS spatial database models for location-based search and route analysis.
- Implemented route optimization and spatial analysis features, and built responsive interfaces for exploring geographic and environmental data.

GIS Developer — State Cartographer's Office, UW–Madison
Madison, WI | *September 2018 – April 2019*

- Built web-based geospatial applications in React and Python for Wisconsin state agencies, including PostGIS analysis pipelines and interactive mapping dashboards for environmental monitoring and public policy research.

EDUCATION

M.S. Geospatial Programming — University of Wisconsin–Madison

September 2017 – August 2019

Geospatial software development, spatial algorithms, network analysis, spatial statistics, and computing.

B.S. Environmental Science / Philosophy (Double Major) — University of Wisconsin–Madison

September 2011 – May 2016
Computational analysis of spatiotemporal datasets. Graduate-level logic, probability theory, and epistemology. Undergraduate publication and conference presentation.

OTHER

- CISSP Training — cybersecurity best practices for production systems.
- Turing Mentor — ongoing mentorship of aspiring software engineers.